

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Science of Living Systems (LS1001)

Time: 1.5 Hours

Semester / Programme: 1st Semester B.Tech

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) What structural feature of membrane phospholipids prevents close packing of fatty acid tails, thereby maintaining plasma membrane fluidity at lower environmental temperatures? [1]

Solution & Key Concept:

- Membrane fluidity is regulated by the saturation level of phospholipid hydrocarbon tails.
- The presence of cis-double bonds in unsaturated fatty acid chains creates kinks or bends in the acyl tails.
- These structural kinks prevent adjacent lipids from packing tightly into a rigid, quasi-crystalline gel state at low temperatures.

Answer: Presence of cis-double bonds (unsaturation) in the fatty acid hydrocarbon chains causing steric hindrance/kinks that hinder compact lipid packing.

Marking Rubric: [1 Mark] for mentioning unsaturated fatty acid tails / cis-double bonds causing kinks preventing tight packing.

1(b) Distinguish between totipotent and pluripotent stem cells in terms of their developmental differentiation potency. [1]

Solution & Key Concept:

- Totipotent stem cells (e.g., zygote up to morula) possess the developmental capacity to differentiate into all embryonic cell lineages as well as extra-embryonic tissues (such as the placenta and umbilical cord).
- Pluripotent stem cells (e.g., inner cell mass of the blastocyst) can differentiate into cells of all three germ layers (ectoderm, mesoderm, and endoderm) but cannot give rise to extra-embryonic supporting structures.

Answer: Totipotent cells can form both embryonic and extra-embryonic tissues; pluripotent cells can form all three embryonic germ layers but not extra-embryonic tissues.

Marking Rubric: [0.5 Mark] for correct definition of totipotency and [0.5 Mark] for distinguishing feature of pluripotency.

1(c) Name the specific catalytic ribosomal component that functions as the peptidyl transferase enzyme during polypeptide chain elongation in prokaryotic protein synthesis. [1]

Solution & Key Concept:

- Peptidyl transferase catalyzes the condensation reaction forming peptide bonds between the aminoacyl-tRNA at the A-site and the peptidyl-tRNA at the P-site.
- In prokaryotes, this reaction is ribozyme-mediated by the catalytic RNA of the large (50S) ribosomal subunit.
- The specific catalytic ribozyme is the 23S ribosomal RNA (23S rRNA).

Answer: 23S rRNA (ribozyme within the 50S ribosomal subunit).

Marking Rubric: [1 Mark] for identifying 23S rRNA (or 23S ribozyme).

1(d) State how the kinetic parameters V and K_m are affected in the presence of a competitive enzyme inhibitor.

[1]

$$K_m^{\text{app}} = K_m \left(1 + \frac{[I]}{K_i} \right), \quad V_{\text{max}}^{\text{app}} = V_{\text{max}}$$

Solution & Key Concept:

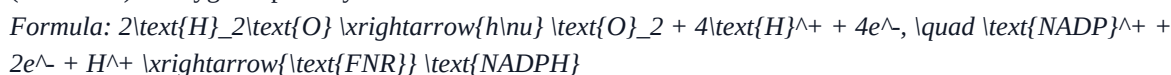
- A competitive inhibitor binds reversibly to the free enzyme at the active site, competing directly with the substrate.
- High concentrations of substrate displace the inhibitor, meaning maximum reaction velocity remains unaffected (V unchanged).
- Because higher substrate concentrations are required to achieve half-maximal velocity, the apparent Michaelis constant increases (K_m increases).

Answer: V remains unchanged; apparent K_m increases.

Marking Rubric: [0.5 Mark] for stating V is unchanged and [0.5 Mark] for stating K_m increases.

1(e) Identify the primary electron donor and the ultimate electron acceptor of non-cyclic electron transport (Z-scheme) in oxygenic photosynthesis.

[1]



Solution & Key Concept:

- In the light reactions of oxygenic photosynthesis, light excitation of photosystem II (P680) causes electron ejection.
- Electrons extracted from water by the oxygen-evolving complex (OEC) regenerate oxidized P680^+ , making H_2O the initial electron donor.
- The electrons are transferred through the photosynthetic electron transport chain to ferredoxin-NADP+ reductase (FNR), where NADP^+ serves as the terminal electron acceptor to form NADPH .

Answer: Primary electron donor: Water (H_2O); Ultimate electron acceptor: NADP^+ .

Marking Rubric: [0.5 Mark] for identifying H_2O and [0.5 Mark] for identifying NADP^+ .

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

- Q.2. (a)** Tabulate three fundamental structural differences between prokaryotic and eukaryotic ribosomes. Explain how these distinctions form the molecular basis for the selective toxicity of broad-spectrum aminoglycoside or macrolide antibiotics. [2.5]

Solution for (a):

- Step 1: Structural differences tabulation:
 - - Overall sedimentation coefficient: Prokaryotes have 70S ribosomes (consisting of 30S small and 50S large subunits); Eukaryotes have 80S cytosolic ribosomes (consisting of 40S small and 60S large subunits).
 - - rRNA Composition: Prokaryotes contain 16S rRNA in the small subunit and 23S + 5S rRNA in the large subunit. Eukaryotes possess 18S rRNA in the small subunit and 28S + 5.8S + 5S rRNA in the large subunit.
 - - Protein count: Prokaryotic 70S ribosome comprises approximately 55 ribosomal proteins; eukaryotic 80S ribosome comprises over 80 proteins.
- Step 2: Basis of Selective Toxicity:
 - - Antibiotics such as aminoglycosides (e.g., streptomycin) selectively bind to the A-site of the bacterial 16S rRNA within the 30S subunit, inducing mRNA codon misreading and mistranslation.
 - - Macrolides (e.g., erythromycin) target the peptidyl transferase center of the bacterial 23S rRNA in the 50S subunit, inhibiting transpeptidation and sterically blocking polypeptide exit.
 - - Because eukaryotic cytoplasmic ribosomes possess distinct 18S/28S rRNA structures, these antibiotic drugs exhibit orders-of-magnitude lower affinity for human 80S ribosomes, sparing host protein synthesis.

Result / Conclusion: Prokaryotes possess 70S (30S+50S) ribosomes with 16S, 23S, and 5S rRNA, whereas eukaryotes have 80S (40S+60S) ribosomes with 18S, 28S, 5.8S, and 5S rRNA. Antibiotics specifically target unique prokaryotic rRNA pockets (e.g., 30S or 50S), selectively killing bacteria without binding eukaryotic 80S ribosomes.

Marking Rubric: [1.5 Marks] for complete tabulation of differences (sedimentation size, subunits, rRNA species), [1 Mark] for explaining selective binding of antibiotics to prokaryotic subunits without affecting host 80S ribosomes.

- (b)** Define cellular homeostasis. Illustrate and describe the essential components of a negative feedback control loop (sensor, control center, effector) taking blood glucose regulation in humans as an engineering control systems analogue. [2.5]

Solution for (b):

- Step 1: Definition of Homeostasis:
 - - Homeostasis is the dynamic physiological self-regulatory process by which living systems maintain a relatively stable internal physical and chemical equilibrium despite external environmental fluctuations.
- Step 2: Negative Feedback Loop Elements & Blood Glucose Control:
 - - 1. Stimulus / Variable: Deviation of blood glucose concentration above or below the physiological set point (approximately 70-100 mg/dL).
 - - 2. Sensor (Receptor): Pancreatic islet cells (Beta-cells detect hyperglycemic deviation via GLUT2 and glucokinase flux; Alpha-cells detect hypoglycemic decline).
 - - 3. Control Center (Integrator): Endocrine pancreas compares the blood glucose level to the biochemical set-point and determines the endocrine response (secretion of insulin or glucagon).
 - - 4. Effector: Insulin acts upon target tissues (liver, skeletal muscle, adipocytes) activating GLUT4 translocation and glycogenesis to decrease blood glucose back to normal; glucagon stimulates hepatic glycogenolysis and gluconeogenesis.
 - - 5. Negative Feedback: Restoration of normoglycemia attenuates further hormonal output, stabilizing the controlled variable.

Result / Conclusion: Homeostasis is dynamic physiological equilibrium maintenance. In negative feedback: Sensor (pancreatic beta/alpha cells) monitors blood glucose; Control Center processes signal and modulates insulin/glucagon release; Effectors (liver/muscle/adipose) execute glucose uptake/storage or release to return glucose levels to the set-point.

Marking Rubric: [1 Mark] for formal definition of homeostasis, [1.5 Marks] for articulating the four architectural components (sensor, integrator, effector, feedback) mapped to blood glucose regulation.

Q.3. (a) Explain the four structural levels of proteins (primary, secondary, tertiary, and quaternary). Detail the specific chemical bonds and interactions that drive and stabilize the tertiary structure of a globular protein.

[3]

Solution for (a):

• Step 1: Four Levels of Protein Architecture:

- - Primary ($1^{\wedge}$) Structure: The linear, covalent sequence of amino acids joined via planar peptide (amide) bonds ($\{-CO-NH-\}$).
- - Secondary ($2^{\wedge}$) Structure: Local spatial folding of the polypeptide backbone stabilized exclusively by hydrogen bonds between carbonyl oxygens ($\{C=O\}$) and amide hydrogens ($\{N-H\}$) of the peptide backbone (e.g., α -helix, β -pleated sheets).
- - Tertiary ($3^{\wedge}$) Structure: The overall comprehensive three-dimensional conformation assumed by a single polypeptide chain, driven by intramolecular interactions between amino acid side-chain (R-group) residues.
- - Quaternary ($4^{\wedge}$) Structure: Spatial arrangement and non-covalent/covalent assembly of two or more distinct folded polypeptide subunits into a functional multi-subunit oligomer (e.g., tetrameric hemoglobin).

• Step 2: Chemical Bonds Stabilizing $3^{\wedge}$ Structure:

- - 1. Hydrophobic Interactions: Non-polar aliphatic/aromatic side chains (e.g., Leu, Ile, Phe) coalesce in the protein interior away from aqueous solvent (driven by water entropy).
- - 2. Hydrogen Bonding: Formed between polar uncharged R-groups (e.g., Ser, Thr, Tyr).
- - 3. Electrostatic / Ionic Interactions (Salt Bridges): Formed between oppositely charged side-chains (e.g., $\{Asp\}^- / \{Glu\}^-$ and $\{Lys\}^+ / \{Arg\}^+$).
- - 4. Van der Waals forces: Weak transient dipole interactions stabilizing closely packed side chains.
- - 5. Disulfide Bonds (Covalent): Covalent cross-links formed via oxidation of thiol groups between two cysteine residues ($\{-S-S-\}$).

Result / Conclusion: Proteins organize into $1^{\wedge}$ (sequence), $2^{\wedge}$ (backbone H-bonds: α -helix/ β -sheet), $3^{\wedge}$ (3D conformation), and $4^{\wedge}$ (multi-subunit association). Tertiary stabilization is driven by hydrophobic collapse, hydrogen bonds, salt bridges, van der Waals forces, and covalent disulfide cross-links.

Marking Rubric: [1.5 Marks] for clearly differentiating the four structural levels, [1.5 Marks] for enumerating and describing the non-covalent interactions and covalent disulfide bridges in tertiary stabilization.

(b) What are Induced Pluripotent Stem Cells (iPSCs)? State any two transcription factors discovered by Shinya Yamanaka used for cellular reprogramming, and discuss one biomedical application of iPSC technology.

[2]

Solution for (b):

• Step 1: Definition of iPSCs:

- - Induced Pluripotent Stem Cells (iPSCs) are genetically reprogrammed adult differentiated somatic cells (such as dermal fibroblasts) that have been retrofitted to an embryonic-like pluripotent state through the ectopic overexpression of defined transcription factor cocktails.

• Step 2: Yamanaka Factors:

- - The canonical Yamanaka reprogramming factors (OSKM) are Oct3/4, Sox2, Klf4, and c-Myc (mentioning any two: e.g., Oct4 and Sox2).

• Step 3: Biomedical Engineering / Therapeutic Application:

- - Patient-Specific Regenerative Medicine: Patient's own somatic cells can be reprogrammed to iPSCs, directed to differentiate into dopaminergic neurons or cardiomyocytes, and transplanted without triggering immune graft rejection.
- - Disease Modeling and Drug Discovery: iPSCs derived from patients harboring genetic pathologies (e.g., ALS or cystic fibrosis) can be differentiated in vitro to screen pharmaceutical drug candidates and assess drug toxicity.

Result / Conclusion: iPSCs are adult somatic cells reprogrammed into a pluripotent state. Key transcription factors include Oct3/4 and Sox2 (or Klf4, c-Myc). Application: autologous cell replacement therapy eliminating immune rejection, or personalized drug screening and genetic disease modeling.

Marking Rubric: [0.5 Mark] for accurate definition of iPSCs, [0.5 Mark] for naming two correct transcription factors, [1 Mark] for describing a valid biomedical engineering/clinical application.

- Q.4. (a)** Given the Michaelis-Menten kinetic expression $v_0 = (V[S] / K_m + [S])$, mathematically derive the Lineweaver-Burk double-reciprocal equation. Explain how the kinetic constants V and K_m are experimentally extracted from the slope, y-intercept, and x-intercept of the resulting plot. [3]

$$\frac{1}{v_0} = \left(\frac{K_m}{V_{\max}} \right) \frac{1}{[S]} + \frac{1}{V_{\max}}$$

Solution for (a):

- Step 1: Mathematical Derivation:
 - - Begin with the Michaelis-Menten equation relating initial velocity (v_0) to substrate concentration ($[S]$):
 - $v_0 = (V[S] / K_m + [S])$
 - - Take the algebraic reciprocal of both sides of the equation:
 - $(1 / v_0) = (K_m / V + [S] / V[S])$
 - - Separate the terms on the right-hand side:
 - $(1 / v_0) = (K_m / V[S]) + ([S] / V[S])$
 - - Simplify to arrive at the Lineweaver-Burk double reciprocal formulation:
 - $(1 / v_0) = (K_m / V)(1 / [S]) + (1 / V)$
 - Step 2: Graphical Parameter Determination:
 - - The equation corresponds to the linear form $y = mx + c$, where $y = (1 / v_0)$ and $x = (1 / [S])$.
 - - y-intercept ($x = 0$): Equals $(1 / V)$, allowing calculation of $V = (1 / \{y\text{-intercept}\})$.
 - - Slope (m): Equals (K_m / V) .
 - - x-intercept ($y = 0$): Setting $(1 / v_0) = 0$ gives $0 = (K_m / V)(1 / [S]) + (1 / V)$, giving $(1 / [S]) = -(1 / K_m)$.
 - - Hence, the x-intercept is $-(1 / K_m)$, giving $K_m = -(1 / \{x\text{-intercept}\})$.
- Result / Conclusion:** Lineweaver-Burk equation: $(1 / v_0) = (K_m / V)(1 / [S]) + (1 / V)$. The y-intercept gives $(1 / V)$, the x-intercept gives $-(1 / K_m)$, and the slope gives (K_m / V) .

Marking Rubric: [1.5 Marks] for step-by-step algebraic reciprocal derivation, [1.5 Marks] for identifying slope, y-intercept, and x-intercept and how V and K_m are calculated.

- (b)** Describe four distinct catalytic strategies used by enzymes to accelerate reaction rates by lowering activation energy (E_a). [2]

Solution for (b):

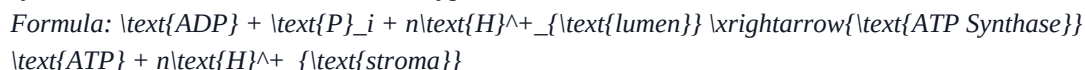
- Enzymes deploy combinations of the following mechanisms to stabilize the transition state and lower the Gibbs free energy of activation:
 1. General Acid-Base Catalysis: Amino acid residues in the active site act as proton donors (acids) or proton acceptors (bases) to stabilize developing charges in the transition state (e.g., His, Glu, Asp).
 2. Covalent Catalysis: The active site contains a reactive nucleophile that forms a transient covalent intermediate with the substrate (e.g., nucleophilic Ser in chymotrypsin).
 3. Metal Ion Catalysis: Coordinated metal ions (e.g., $\{Zn\}^{2+}$, $\{Mg\}^{2+}$) facilitate substrate binding, polarize bonds, generate nucleophiles by deprotonating water, or participate in reversible redox processes.
 4. Catalysis by Approximation and Orientation: The enzyme binds substrates in close proximity and with precise geometric alignment optimal for reaction, drastically reducing the entropic penalty of bimolecular reactions.

Result / Conclusion: 1. Acid-Base Catalysis (proton transfers); 2. Covalent Catalysis (transient covalent bond); 3. Metal Ion Catalysis (electrophilic catalysis / charge stabilization); 4. Approximation and Orientation (proper alignment reducing entropy).

Marking Rubric: [0.5 Mark] each for naming and concisely defining four catalytic strategies.

Q.5. (a) Outline the path of electron transport in the photosynthetic Z-scheme from Photosystem II (PSII) to Photosystem I (PSI). Explain how this electron flow couples with proton translocation to drive ATP synthesis via Mitchell's chemiosmotic hypothesis.

[3]



Solution for (a):

• Step 1: Z-Scheme Electron Transport Pathway:

- - Photons excite the reaction center P680 of PSII. Excited $P680^*$ transfers electrons to Pheophytin (*Pheo*).
- - Electrons pass sequentially through plastoquinones: $Pheo \rightarrow Q_A \rightarrow Q_B$. Upon two-electron reduction and stroma protonation, Q_B becomes plastoquinol ($\{PQH\}_2$).
- - $\{PQH\}_2$ diffuses within the thylakoid lipid bilayer to Cytochrome b_{6f} complex, where electrons flow through the Q-cycle to the mobile copper protein plastocyanin (*PC*).
- - Plastocyanin transfers electrons to reduce the photo-oxidized reaction center $P700^+$ of Photosystem I (PSI).
- - Photo-excitation of PSI transfers electrons via A_0 (chlorophyll), A_1 (phylloquinone), and iron-sulfur clusters (F_X , F_A , F_B) to Ferredoxin (*Fd*), which reduces $\{NADP\}^+$ to $\{NADPH\}$ via FNR.

• Step 2: Generation of Proton Gradient & Chemiosmosis:

- - Protons accumulate in the thylakoid lumen via: (i) Photolysis of water ($2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4e^-$) by the oxygen-evolving complex (OEC), and (ii) Proton pumping from stroma to lumen by the Cytochrome b_{6f} complex via the plastoquinone redox loop.
- - This creates a steep proton electrochemical gradient across the thylakoid membrane ($\Delta \text{pH} \approx 3.5$ units).
- - Protons flow exergonically down their concentration and electrical gradient back into the stroma through the $CF_0\text{-}CF_1$ ATP Synthase complex, driving rotary mechanical catalysis to synthesize ATP from $\{ADP\}$ and inorganic phosphate ($\{P\}_i$).

Result / Conclusion: Electrons flow: $\{H\}_{2O} \rightarrow \{P680\} \rightarrow \{Pheo\} \rightarrow Q_A \rightarrow Q_B \rightarrow \{Cyt\} b_{6f} \rightarrow \{Plastocyanin\} \rightarrow \{P700\} \rightarrow Fd \rightarrow \{FNR\} \rightarrow \{NADP\}^+$. Proton translocation into the lumen creates a proton-motive force that drives ATP generation via ATP synthase.

Marking Rubric: [1.5 Marks] for the complete sequence of electron carriers in the Z-scheme, [1.5 Marks] for describing proton accumulation mechanisms and ATP generation via chemiosmosis through ATP synthase.

(b) Summarize the three sequential stages of the Calvin Cycle (Dark Reaction) in C_3 photosynthesis. Write the net stoichiometric equation showing the input of $\{CO\}_2$, ATP, and NADPH required to synthesize one molecule of glucose.

[2]

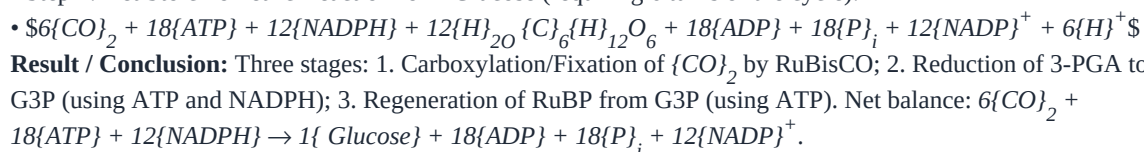


Solution for (b):

• Step 1: Three Stages of the Calvin Cycle:

- 1. Carbon Fixation: Ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) catalyzes the carboxylation of ribulose-1,5-bisphosphate (RuBP, 5C) with $\{CO\}_2$ to yield an unstable 6C intermediate that cleaves into two molecules of 3-phosphoglycerate (3-PGA, 3C).
- 2. Reduction Phase: 3-PGA is phosphorylated by ATP (generating 1,3-bisphosphoglycerate) and subsequently reduced by NADPH to form glyceraldehyde-3-phosphate (G3P / GAP, a 3C triose phosphate).
- 3. Regeneration of RuBP: Through complex sugar-phosphate rearrangements consuming ATP, 5 out of every 6 molecules of G3P are recycled to regenerate RuBP (the $\{CO\}_2$ acceptor), allowing cycle continuation. The remaining net triose exits to synthesize glucose.

• Step 2: Net Stoichiometric Reaction for 1 Glucose (requiring 6 turns of the cycle):



Marking Rubric: [1 Mark] for describing the 3 stages with enzyme and intermediate names, [1 Mark] for the correct balanced stoichiometric equation for one mole of glucose.

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